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COURSE: **Design and Analysis of Algorithms**

CODE: **CSA0619**

REPORT

Q23. A hospital system stores emergency patient data unsorted for quick updates.

Doctors need to locate records rapidly.

a) Explain how Linear Search helps in this scenario.

b) Analyze its performance in different cases.

c) Justify why sorting may not always be preferred.

Source Code (Python Example)

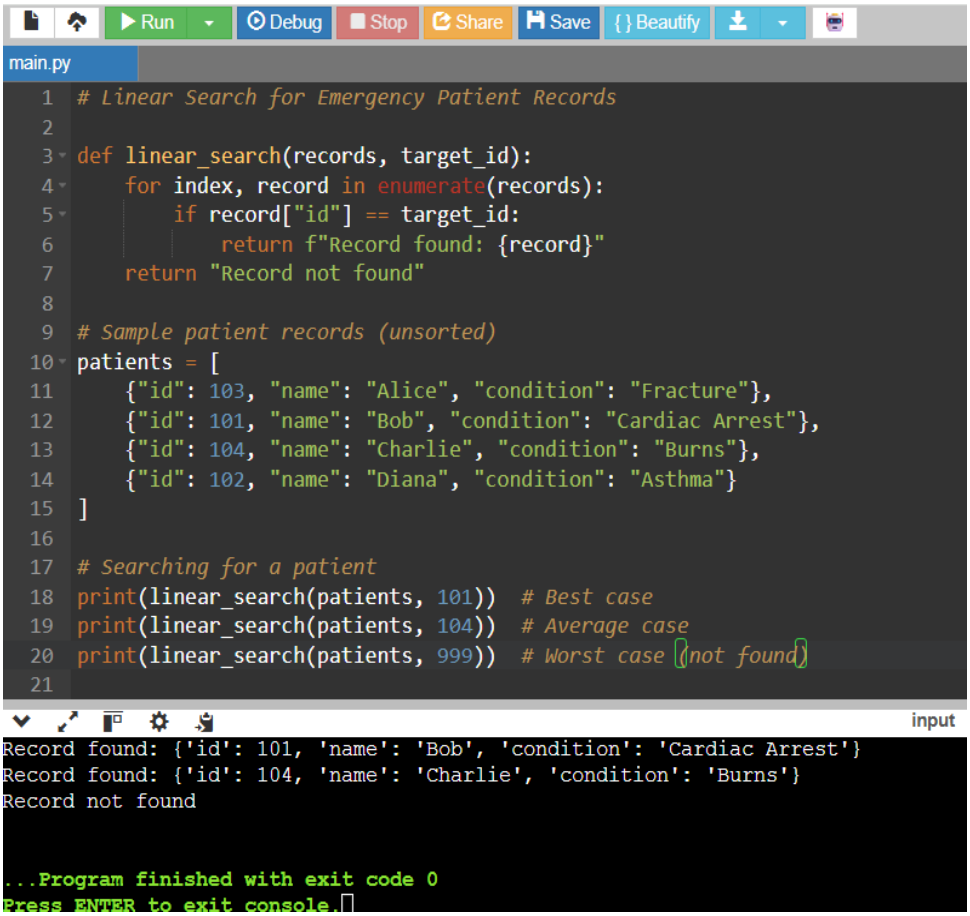
```
# Linear Search for Emergency Patient Records

def linear_search(records, target_id):
    for index, record in enumerate(records):
        if record["id"] == target_id:
            return f"Record found: {record}"
    return "Record not found"

# Sample patient records (unsorted)
patients = [
    {"id": 103, "name": "Alice", "condition": "Fracture"},
    {"id": 101, "name": "Bob", "condition": "Cardiac Arrest"},
    {"id": 104, "name": "Charlie", "condition": "Burns"},
    {"id": 102, "name": "Diana", "condition": "Asthma"}
]

# Searching for a patient
print(linear_search(patients, 101))  # Best case
print(linear_search(patients, 104))  # Average case
```

print(linear_search(patients, 999)) # Worst case (not found)



a) **How Linear Search Helps**

- **Unsorted Data Access**: In emergencies, patient records are updated frequently. Linear search allows doctors to quickly scan through unsorted data without needing preprocessing.
- **Simplicity**: It works directly on the dataset without requiring additional structures.
- **Flexibility**: Records can be inserted or updated instantly without worrying about maintaining order.

b) Performance Analysis

- **Best Case**: Patient record is the first entry → **O(1)** time.
- **Worst Case**: Patient record is the last entry or not present → **O(n)** time.
- **Average Case**: On average, the search will take about **n/2 comparisons**, which is still **O(n)**.

c) Why Sorting May Not Always Be Preferred

- **Time Overhead**: Sorting after every update is costly (O(n log n)).
- **Dynamic Updates**: Emergency records change rapidly; maintaining sorted order slows down insertion.
- **Real-Time Needs**: Doctors prioritize immediate access over optimized search.

Case	Description	Time Complexity
Best	Record is the first entry	O(1)
Average	Record is somewhere in the middle	O(n/2) ≈ O(n)
Worst	Record is last or not present	O(n)

Linear Search is a practical solution for hospital emergency systems where:

- Data is unsorted due to frequent updates.
- Doctors require immediate access to patient records.
- Sorting introduces unnecessary delays.

While not the most efficient algorithm for large datasets, Linear Search balances **speed, flexibility, and simplicity**, making it suitable for real-time healthcare applications.